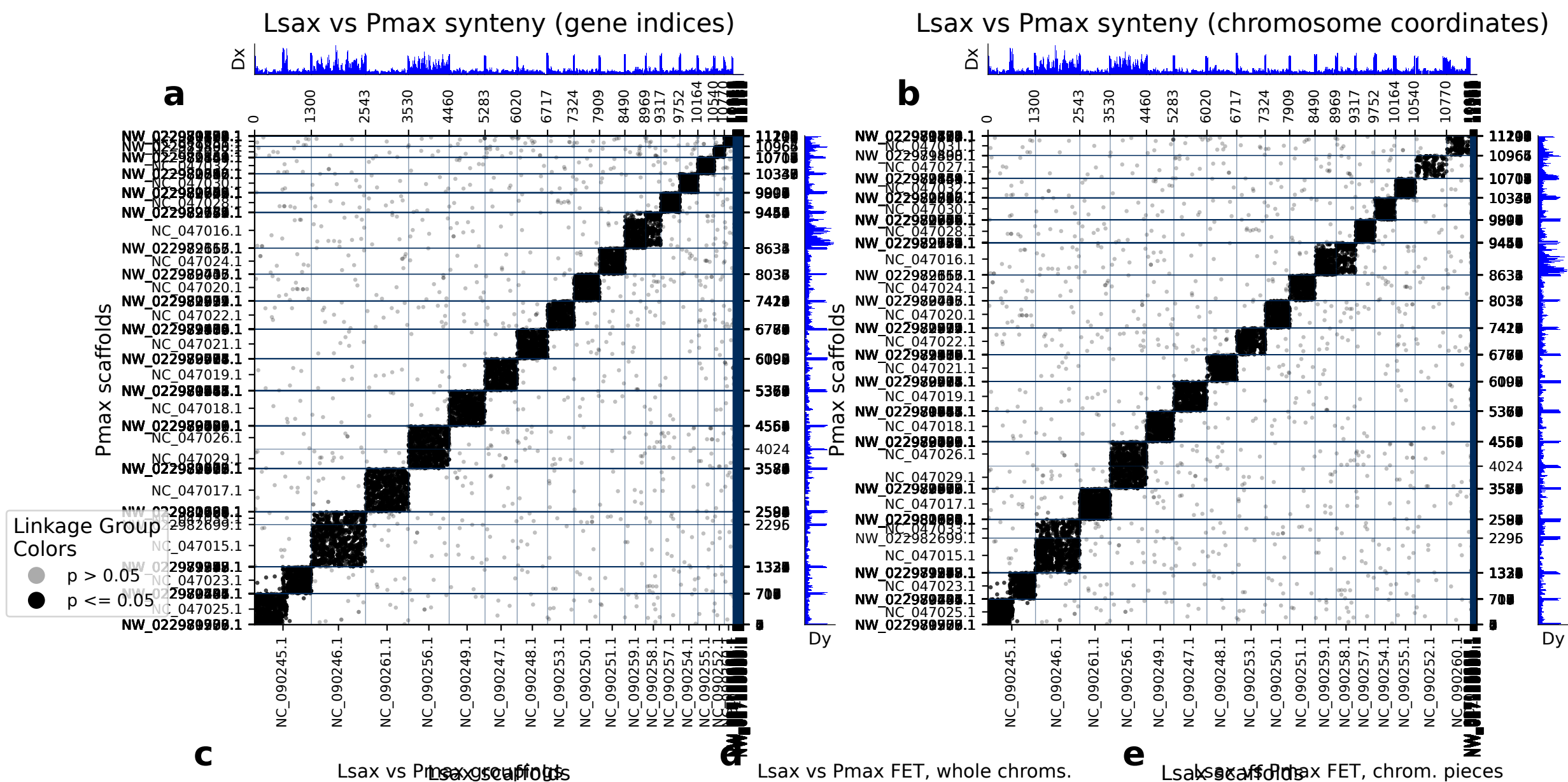


diamond: 11213 orthologs on 232 Lsax scaffolds and 192 Pmax scaffolds.



	Lsax scaf	Pmax scaf	p (FET)	Count
0	NC_090261.1	NC_047017.1	0.00e+00	903
1	NC_090246.1	NC_047015.1	0.00e+00	889
2	NC_090249.1	NC_047018.1	0.00e+00	741
3	NC_090247.1	NC_047019.1	0.00e+00	660
4	NC_090245.1	NC_047025.1	0.00e+00	632
5	NC_090248.1	NC_047021.1	0.00e+00	627
6	NC_090245.1	NC_047023.1	0.00e+00	546
7	NC_090253.1	NC_047022.1	0.00e+00	539
8	NC_090251.1	NC_047024.1	0.00e+00	529
9	NC_090250.1	NC_047020.1	0.00e+00	528
10	NC_090256.1	NC_047026.1	0.00e+00	466
11	NC_090259.1	NC_047016.1	0.00e+00	424
12	NC_090256.1	NC_047029.1	0.00e+00	381
13	NC_090257.1	NC_047028.1	0.00e+00	379
14	NC_090254.1	NC_047030.1	0.00e+00	361
15	NC_090255.1	NC_047032.1	0.00e+00	310
16	NC_090258.1	NC_047016.1	6.52e-286	290
17	NC_090246.1	NC_047033.1	6.48e-194	247
18	NC_090252.1	NC_047027.1	7.35e-294	184
19	NC_090260.1	NC_047031.1	9.71e-239	152
20	NC_030595.1	NW_022980724.1	1.90e-07	3
21	NW_027125817.1	NW_022980858.1	1.28e-02	2

	Lsax scaf	Pmax scaf	p (FET)	Count
0	NC_090261.1 : 0-end	NC_047017.1 : 0-end	0.00e+00903	
1	NC_090246.1 : 0-end	NC_047015.1 : 0-end	0.00e+00889	
2	NC_090249.1 : 0-end	NC_047018.1 : 0-end	0.00e+00741	
3	NC_090247.1 : 0-end	NC_047019.1 : 0-end	0.00e+00660	
4	NC_090245.1 : 0-end	NC_047025.1 : 0-end	0.00e+00632	
5	NC_090248.1 : 0-end	NC_047021.1 : 0-end	0.00e+00627	
6	NC_090245.1 : 0-end	NC_047023.1 : 0-end	0.00e+00546	
7	NC_090253.1 : 0-end	NC_047022.1 : 0-end	0.00e+00539	
8	NC_090251.1 : 0-end	NC_047024.1 : 0-end	0.00e+00529	
9	NC_090250.1 : 0-end	NC_047020.1 : 0-end	0.00e+00528	
10	NC_090256.1 : 0-end	NC_047026.1 : 0-end	0.00e+00466	
11	NC_090259.1 : 0-end	NC_047016.1 : 0-end	0.00e+00424	
12	NC_090256.1 : 0-end	NC_047029.1 : 0-end	0.00e+00381	
13	NC_090257.1 : 0-end	NC_047028.1 : 0-end	0.00e+00379	
14	NC_090254.1 : 0-end	NC_047030.1 : 0-end	0.00e+00361	
15	NC_090255.1 : 0-end	NC_047032.1 : 0-end	0.00e+00310	
16	NC_090258.1 : 0-end	NC_047016.1 : 0-end	6.52e-286290	
17	NC_090246.1 : 0-end	NC_047033.1 : 0-end	6.48e-194247	
18	NC_090252.1 : 0-end	NC_047027.1 : 0-end	7.35e-294184	
19	NC_090260.1 : 0-end	NC_047031.1 : 0-end	9.71e-239152	
20	NC_030595.1 : 0-end	NW_0229808724.1 : 0-end	9.0e-07 3	
21	NW_027125817.1 : 0-end	NW_022980858.1 : 0-end	2.8e-02 2	

(a) depicts the chromosomes/scaffolds of Lsax (x-axis) plotted against the chromosomes/scaffolds of Pmax (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 11213 orthologs found between 232 Lsax scaffolds and 192 Pmax scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. *Nature Ecology & Evolution* 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.